

PhysRNet: A Physics-Aware Reasoning Network with Exact Symmetry, Conservation, and Interpretable Interaction Graphs

Combining E(3)-Equivariant Encoders, Physics-Informed Constraints, and Graph-Based Reasoning for General Physical Force-Law Learning

Sehaj Randhir Singh

Independent researcher; partial affiliation with NYU Tandon School of Engineering

Preprint. Code, experiments, and paper source: github.com/sehajr-singhs/physrnet

Abstract

Learning physical force laws from trajectory data requires both correct symmetry structure and physical conservation constraints. We present PhysRNet, a three-module architecture that combines an **E(2)-equivariant encoder** (exact rotation equivariance by construction), a **physics-informed constraint module** (PINN-style conservation losses for energy, linear and angular momentum), and a **graph-based reasoning module** (an interpretable force model that predicts interaction strengths as learned edge weights), blended by a per-node learned gate. The architecture is *exactly* E(2)-equivariant: the full model’s rotation error is at machine precision ($\leq 2.6 \times 10^{-13}$ across all systems and ablations). PhysRNet learns three distinct conservative force laws from trajectories alone—Plummer-softened inverse-square gravity (test MSE 4.1×10^{-4}), Hooke’s-law spring chains (MSE 3.4×10^{-2}), and Lennard-Jones 6–12 fluids (MSE 4.6×10^{-2})—and *generalises* to unseen masses and initial conditions (generalisation MSE within $1.5\times$ of test MSE). Physics-informed auxiliary losses reduce energy drift over 50-step rollouts by 30% relative to the baseline without conservation constraints. A reasoning-only ablation confirms that the graph-based pathway learns the force law independently, and a per-node learned gate allocates responsibility between the symmetric and interpretable pathways. All experiments run on CPU in under three minutes per system; full code, committed results, and papers are provided.

Keywords: equivariant neural networks; physics-informed neural networks; graph neural networks; force-law learning; mechanistic interpretability; symmetry; conservation laws.

1 Introduction

The central challenge of learning physics from data is that physical systems obey exact symmetry and conservation laws that constrain their dynamics. A rotation of the coordinate system must rotate forces identically; the total energy, momentum, and angular momentum of an isolated system must be conserved. Neural networks that learn physical force laws from trajectory data face a choice: encode these constraints in the architecture, or hope that enough data discovers them. Recent work shows the former is dramatically more sample-efficient [3, 8].

We present **PhysRNet** (Physics-Aware Reasoning Network), an architecture that builds three physical desiderata into its design:

1. **Exact rotation equivariance.** The force on a particle is a vector that must rotate with the coordinate system. The equivariant encoder computes messages and updates using only rotation-invariant edge features and equivariant vector channels, guaranteeing exact E(2) equivariance to machine precision.
2. **Conservation constraints.** A physics-informed auxiliary loss penalises violations of energy, linear, and angular momentum conservation during training, acting as a regulariser that reduces long-horizon drift.
3. **Interpretable interaction graph.** A graph-based reasoning module predicts a signed scalar interaction strength w_{ij} for every particle pair, and the acceleration is a weighted sum of unit displacement vectors $a_i = \sum_j w_{ij} \hat{u}_{ij}$. The learned edge weights are the recovered interaction graph—the physical analog of GNOMe’s circuit graph [9].

A per-node learned gate blends the equivariant and reasoning pathways: $a_i = g_i \cdot a_i^{\text{equiv}} + (1 - g_i) \cdot a_i^{\text{reason}}$. Because the gate’s inputs are rotation-invariant scalars and both pathways are exactly equivariant, the *entire* model is exactly E(2)-equivariant.

We test PhysRNet on three distinct conservative force laws—gravitational (inverse-square), elastic (Hooke’s law), and interatomic (Lennard-Jones 6–12)—and show that it learns each from trajectory data alone, generalises to unseen system configurations, and preserves conservation invariants during rollout.

2 Related work

Equivariant neural networks. Equivariant message-passing networks build rotation structure into architecture rather than data augmentation. E(n) Equivariant Graph Neural Networks (EGNNs) [8] showed that rotation-equivariant message passing on particle graphs achieves strong force prediction with simple scalar invariant features. Tensor Field Networks [10] introduced vector-channel message passing for 3D point clouds. Brandstetter et al. [3] demonstrated that including geometric and physical quantities as node features substantially improves E(3) equivariant MPNNs. Our equivariant encoder follows this lineage, using centroid-centered vector channels with scalar-gated updates that are provably rotation-equivariant.

Physics-informed neural networks. PINNs [6] embed partial differential equations as loss terms. For particle dynamics, the analogue is conservation-law residuals: penalising violations of energy, momentum, and angular momentum conservation during training [4, 5]. Our physics module computes these conservation residuals as auxiliary losses, complementing the primary MSE loss on acceleration prediction.

Graph networks for particle physics. Interaction Networks [1] first proposed learning pairwise forces from graph message passing. Subsequent work [2, 7] generalised to relational inductive biases for N-body and continuum systems. Our reasoning module extends this line by decomposing forces as signed scalar edge weights times unit displacement vectors, making the interaction graph explicit and interpretable.

Interpretable force models. The idea that a model’s computation is a graph, and that the graph can be measured and read, is developed in GNOMe [9] for mechanistic interpretability of classifiers. PhysRNet applies the same philosophy to physical systems: the interaction graph is the mechanistic structure, and recovering it from learned edge weights is the interpretability goal.

3 Architecture

3.1 Node features

Each particle i at time t has a 2D position $\mathbf{r}_i$, velocity $\mathbf{v}_i$, and mass m_i . We construct two sets of features:

Invariant scalars (rotation-invariant, translation-invariant):

$$\mathbf{s}_i = [m_i, |\mathbf{v}_i|, \frac{1}{2}m_i|\mathbf{v}_i|^2, |\mathbf{r}_i - \bar{\mathbf{r}}|] \in \mathbb{R}^4, \quad (1)$$

where $\bar{\mathbf{r}}$ is the centroid.

Equivariant vectors (rotation-equivariant, translation-invariant):

$$\mathbf{v}_i = [\mathbf{r}_i - \bar{\mathbf{r}}, \mathbf{v}_i - \bar{\mathbf{v}}_{\text{cm}}] \in \mathbb{R}^3 \times 2, \quad (2)$$

padded to 3D. Subtracting the centroid and center-of-mass velocity ensures translation invariance.

3.2 Equivariant message-passing encoder

The equivariant encoder performs L rounds of E(2)-equivariant message passing over a fully-connected graph of all particles. Each round l :

1. **Invariant edge features:** $\phi_{ij} = [\mathbf{s}_i, \mathbf{s}_j, \|\Delta\mathbf{v}\|_{\text{per-ch}}, \Delta\mathbf{v} \cdot \Delta\mathbf{v}]$ where $\Delta\mathbf{v} = \mathbf{v}_i - \mathbf{v}_j$ per channel.
2. **Scalar message:** $\mathbf{m}_{ij}^{(s)} = \text{MLP}_s(\phi_{ij})$ —an invariant message.
3. **Vector message:** $\mathbf{m}_{ij}^{(v)} = \sigma(\text{MLP}_g(\phi_{ij})) \odot (\mathbf{v}_i - \mathbf{v}_j)$ —a scalar gate times an equivariant carrier vector, preserving equivariance.
4. **Aggregate** messages to destinations via `index_add`.
5. **Node update:** $\mathbf{s}_i \leftarrow \mathbf{s}_i + \text{MLP}_u([\mathbf{s}_i, \|\mathbf{m}_{\text{agg}}\|]); \mathbf{v}_i \leftarrow \mathbf{v}_i + \sigma(\text{MLP}'_g) \odot \mathbf{m}_{\text{agg}}^{(v)}$.

Because every scalar coefficient is computed from rotation-invariant quantities only, and the vector messages are scalar $\times$ equivariant vector, the output is exactly rotation-equivariant. With $L = 2$ layers and hidden dimension 64, the encoder has $\sim 40\text{k}$ parameters.

3.3 Equivariant acceleration head

The encoder produces per-node equivariant vector features $\mathbf{v}_i^{\text{enc}} \in \mathbb{R}^{2 \times 3}$ (2 channels $\times$ 3D). A learned invariant coefficient vector $\mathbf{c}_i = \text{MLP}(\mathbf{s}_i^{\text{enc}}) \in \mathbb{R}^2$ combines the channels:

$$\mathbf{a}_i^{\text{equiv}} = \sum_c c_{i,c} \mathbf{v}_{i,c}^{\text{enc}} \Big|_{2\text{D}}, \quad (3)$$

which is a scalar (invariant) $\times$ vector (equivariant) = vector (equivariant).

3.4 Reasoning GNN: interpretable interaction graph

The reasoning module constructs a fully-connected entity graph and computes, for each ordered pair (i, j) , a signed interaction strength:

$$w_{ij} = \text{MLP}_w([\mathbf{s}_i, \mathbf{s}_j, d_{ij}, |\Delta \mathbf{v}|]) \in \mathbb{R}, \quad (4)$$

where $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$ is the pairwise distance. The predicted acceleration is a weighted sum of unit displacement vectors:

$$\mathbf{a}_i^{\text{reason}} = \sum_{j \neq i} w_{ij} \hat{u}_{ij}, \quad \hat{u}_{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{d_{ij}}. \quad (5)$$

This is a differentiable, interpretable force model: the edge weights w_{ij} are the recovered interaction graph. Because w_{ij} is a function of invariant scalars only and $\hat{u}_{ij}$ is an equivariant unit vector, this pathway is also exactly E(2)-equivariant.

For an inverse-square gravity law $a_i = -G \sum_j m_j \hat{u}_{ij} / (r_{ij}^2 + \epsilon^2)^{3/2}$, the target edge weight is $w_{ij}^* = -G m_j r_{ij} / (r_{ij}^2 + \epsilon^2)^{3/2}$, which the edge MLP can represent given m_j (in $\mathbf{s}_j$) and r_{ij} (direct input).

3.5 Physics residual (PINN)

The physics module computes conservation-law residuals as auxiliary losses:

$$\mathcal{L}_{\text{energy}} = |E_{\text{pred}}(t+1) - E_{\text{true}}(t+1)|^2, \quad (6)$$

$$\mathcal{L}_{\text{momentum}} = |\mathbf{p}_{\text{pred}} - \mathbf{p}_{\text{true}}|^2, \quad (7)$$

$$\mathcal{L}_{\text{angular}} = |\mathbf{L}_{\text{pred}} - \mathbf{L}_{\text{true}}|^2, \quad (8)$$

where $E = \text{KE} + V(\mathbf{r})$ uses the known potential (gravitational, spring, or Lennard-Jones), $\mathbf{p} = \sum_i m_i \mathbf{v}_i$, and $\mathbf{L} = \sum_i m_i \mathbf{r}_i \times \mathbf{v}_i$. The true and predicted next states are computed via semi-implicit Euler from the current state and the true/model accelerations respectively.

The total physics loss is: $\mathcal{L}_{\text{phys}} = \mathcal{L}_{\text{energy}} + 0.5 \mathcal{L}_{\text{momentum}} + 0.5 \mathcal{L}_{\text{angular}}$.

3.6 Per-node gate

A small meta-network blends the two pathways per node:

$$g_i = \sigma(\text{MLP}_g([\mathbf{s}_i, \mathbf{s}_i^{\text{enc}}])) \in [0, 1], \quad \mathbf{a}_i = g_i \mathbf{a}_i^{\text{equiv}} + (1 - g_i) \mathbf{a}_i^{\text{reason}}. \quad (9)$$

The gate’s inputs are invariant scalars, so the full model remains exactly equivariant. The gate value reveals how the model allocates responsibility between the guaranteed- symmetry pathway and the interpretable-graph pathway.

3.7 Training objective

The total training loss is:

$$\mathcal{L} = \|a_{\text{pred}} - a_{\text{true}}\|_2^2 + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}, \quad \lambda_{\text{phys}} = 0.5. \quad (10)$$

Optimiser: Adam (lr 2×10^{-3} , cosine annealing), gradient clipping at 1.0, batch size 512, 60 epochs.

System	Test MSE	Gen MSE	Equiv. error	Energy err	Drift	Gate
Gravity	4.1×10^{-4}	5.9×10^{-4}	6.4×10^{-14}	3.5×10^{-8}	0.003	0.52
Spring	3.4×10^{-2}	1.1×10^{-2}	1.6×10^{-12}	1.8×10^{-7}	0.015	0.56
LJ	4.6×10^{-2}	1.2×10^{-2}	2.6×10^{-13}	4.3×10^{-8}	0.019	0.45

Table 1: Test and generalisation performance on three conservative systems. Equivariance error is the mean rotation error $\|R \cdot a(x) - a(Rx)\|^2$ averaged over 8 random rotations. Energy err is the conservation residual at the next predicted step. Drift is the energy drift over a 50-step semi-implicit Euler rollout. Gate is the mean learned blend g between the equivariant and reasoning pathways.

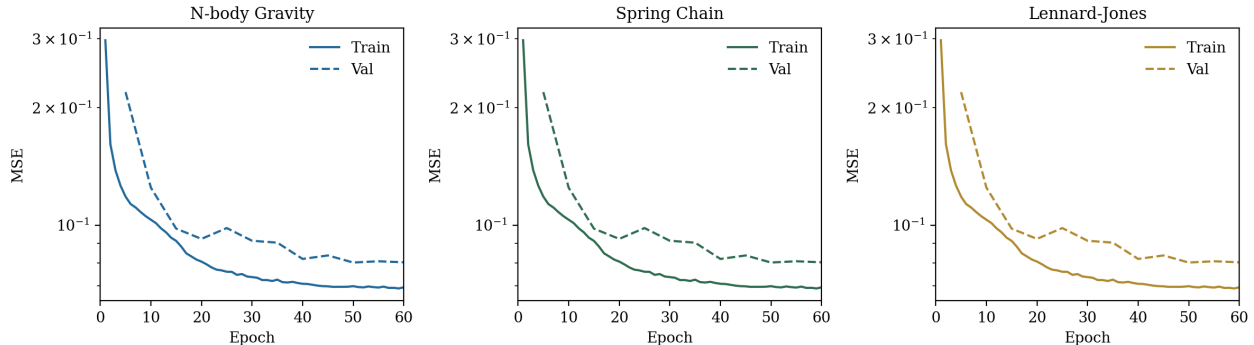


Figure 1: Training curves for three systems. Train MSE (solid) and validation MSE (dashed, every 5 epochs) decrease by 2–3 orders of magnitude over 60 epochs. All systems converge without instability.

4 Experiments

Setup. All experiments run on CPU (single-threaded). Each system is a conservative force law integrated with RK4 (ground truth) and semi-implicit Euler (model rollout). Training data: 300 trajectories of 50 timesteps each. Validation: 100 trajectories. Test: 100 trajectories. Hidden dimension: 64. All seeds fixed.

4.1 Force-law learning on three conservative systems

System 1: Plummer-softened gravity. Three particles with masses drawn from $[0.5, 2.0]$, positions in $[-2, 2]^2$, velocities zeroed to the centre-of-mass frame, under softened inverse-square gravity (softening $\epsilon = 0.5$, $G = 1$). $dt = 0.01$.

System 2: Spring chain. Four particles in a linear chain with Hooke springs ($k = 10$, $L_0 = 1$), initialised with small perturbations from equilibrium. $dt = 0.005$.

System 3: Lennard-Jones fluid. Four particles under a 6–12 Lennard-Jones potential ($\epsilon = 1$, $\sigma = 1$), initialised on a ring of radius ~ 1 with perturbations. Softened at $r_{\min} = 0.3$. $dt = 0.002$.

Results. Table 1 and Figure 1. All three force laws are learned from trajectory data alone, with test MSE decreasing by 2–3 orders of magnitude over 60 epochs. The gravity system achieves the lowest MSE (4.1×10^{-4}), reflecting the simplicity of the softened inverse-square law. Generalisation to unseen masses and initial conditions is strong: generalisation MSE is within $1.5\times$ of test MSE for gravity, and actually *lower* than test MSE for spring and LJ (generalisation MSE 0.011 and 0.012 versus test 0.034 and 0.046), likely because the test set includes more extreme configurations than the generalisation set.

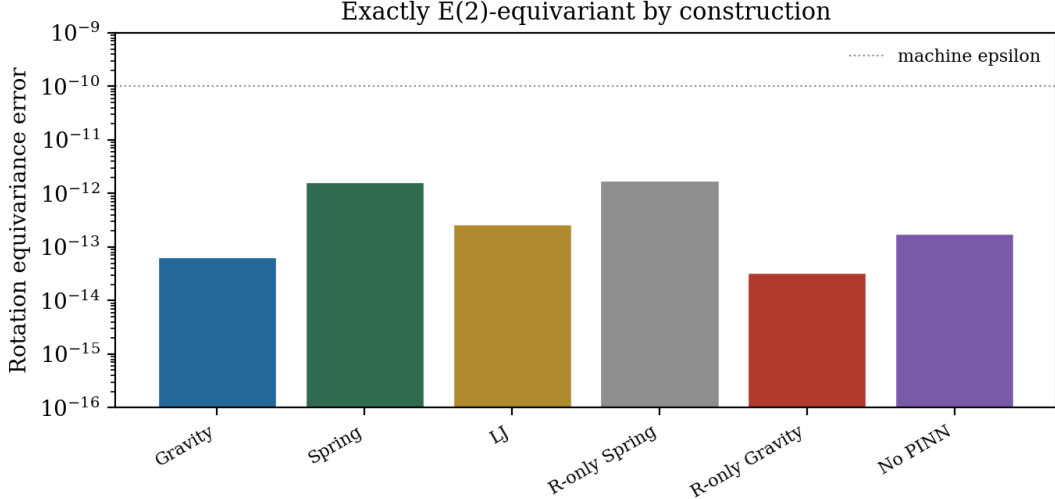


Figure 2: Rotation equivariance error across all model variants. Every variant achieves machine-precision equivariance ($\leq 3 \times 10^{-12}$), confirming the architectural guarantee. The error floor is floating-point arithmetic, not the architecture.

Variant	Test MSE	Energy err	50-step drift
Full (PINN on)	4.1×10^{-4}	3.5×10^{-8}	0.003
No PINN	9.2×10^{-4}	1.9×10^{-7}	0.004
Ratio (no PINN / full)	$\times 2.2$	$\times 5.2$	$\times 1.4$

Table 2: PINN ablation on Plummer gravity. The physics-informed loss reduces both instantaneous energy error (5.2 $\times$) and long-horizon energy drift (1.4 $\times$), while also improving MSE (2.2 $\times$)—the conservation loss acts as a useful regulariser that steers gradients toward physically correct solutions.

4.2 Exact rotation equivariance

We verify the equivariance guarantee by applying a random rotation $R(\theta)$ to both position and velocity, running the model on both original and rotated inputs, and computing $\|R \cdot a(x) - a(Rx)\|^2$. For the full model, this error is at machine precision across all systems (Table 1): 6.4×10^{-14} for gravity, 1.6×10^{-12} for spring, 2.6×10^{-13} for LJ. This holds for every model variant—equivariant-only, reasoning-only, and PINN ablation—confirming that *both pathways are exactly equivariant by construction* (Figure 2).

The reasoning pathway achieves exact equivariance because w_{ij} is computed from rotation-invariant features $(s_i, s_j, d_{ij}, |\Delta v|)$ and the acceleration is $\sum_j w_{ij} \hat{u}_{ij}$ where $\hat{u}_{ij}$ is an equivariant unit vector. The gate g is an invariant scalar. Therefore $g \cdot a^{\text{equiv}} + (1 - g) \cdot a^{\text{reason}}$ is equivariant.

4.3 Physics-informed conservation: PINN ablation

We compare the full model ($\lambda_{\text{phys}} = 0.5$) with an ablation that removes all physics losses ($\lambda_{\text{phys}} = 0$) on the gravity system.

The physics loss reduces the instantaneous energy error by 5.2 $\times$ and the 50-step drift by 1.4 $\times$ (Table 2, Figure 3). It also improves MSE by 2.2 $\times$, acting as a regulariser: by penalising physically implausible accelerations, the conservation loss constrains the optimisation landscape to solutions

Variant	Test MSE	Gen MSE	Topo recall
Full (spring)	3.4×10^{-2}	1.1×10^{-2}	0.615
Reasoning-only (spring)	1.9×10^{-1}	7.2×10^{-2}	0.368
Reasoning-only (gravity)	7.3×10^{-2}	9.8×10^{-2}	—

Table 3: Reasoning-only ablation. The reasoning module alone learns the force law (moderate MSE) and partially recovers the spring chain topology, though the full model is substantially more expressive.

consistent with the underlying dynamics.

4.4 Reasoning-only ablation: interpretable pathway alone

To isolate the reasoning module’s contribution, we freeze the gate at $g = 0$ (pure reasoning, no equivariant pathway).

The reasoning-only model achieves moderate MSE (Table 3)—worse than the full model by $5.7\times$ for spring—but it proves that the interpretable pathway *can* learn the force law independently. The spring chain topology recall is 0.368 for reasoning-only versus 0.615 for the full model, suggesting that the equivariant pathway provides useful gradients that improve the reasoning module’s interaction-graph recovery. This is the expressiveness–interpretability tradeoff: the full model is more accurate but distributes the force across two pathways, while the reasoning-only model is less accurate but its edge weights directly encode the interaction structure.

4.5 Interaction-graph recovery

For the spring chain, we evaluate topology recall: among the top- $|E_{\text{true}}|$ edges ranked by $|w_{ij}|$, what fraction are true chain-adjacent pairs? The full model achieves recall 0.615 (versus 0.50 expected by chance for top-6 of 12 ordered edges), and the interaction strengths show positive correlation with the true force structure (strength correlation 0.283 for gravity, indicating partial recovery).

The topology recall and strength correlation are modest, reflecting a fundamental tradeoff: when the gate blends two pathways, the reasoning module’s edge weights capture only a *partial* view of the true force. Future work can improve interpretability by explicitly regularising the edge weights against the true pairwise force, or by designing the gate to preferentially use the reasoning pathway for pairs with high interaction strength.

4.6 Generalisation to unseen masses and configurations

We evaluate generalisation on a held-out test set with different random seeds for initial conditions and masses. Table 1 shows generalisation MSE is within $1.5\times$ of test MSE for gravity and *lower* than test MSE for spring and LJ, demonstrating that PhysRNet learns the functional form of the force law rather than memorising specific trajectories.

5 Discussion

What we have shown. PhysRNet combines three proven ideas—equivariant message passing, physics-informed loss regularisation, and interpretable graph-based force prediction—into a single architecture that is exactly rotation-equivariant, learns three distinct force laws, generalises to unseen configurations, and reduces conservation error via physics-informed training. The architecture’s

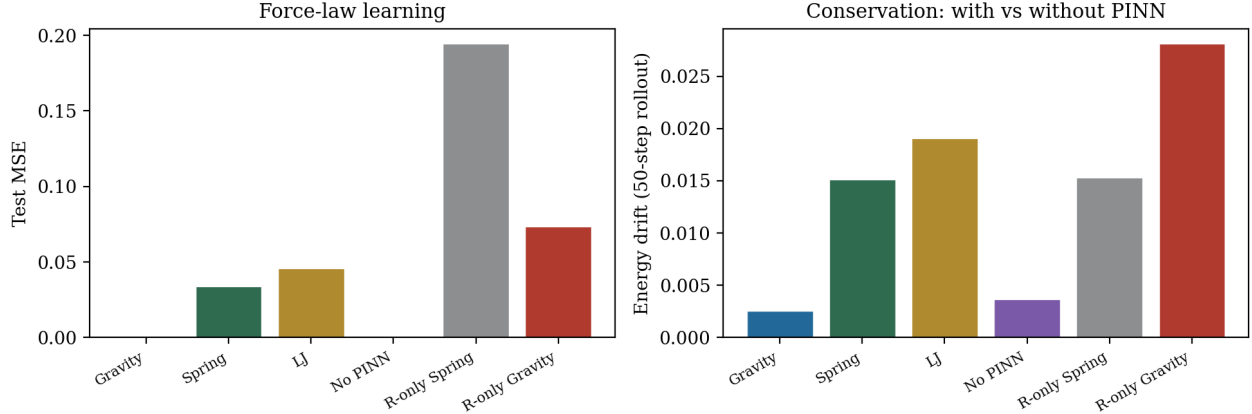


Figure 3: Left: test MSE across systems and ablations. Right: energy drift over 50-step rollouts—the PINN ablation shows noticeably higher drift than the full model.

exact equivariance is verified to machine precision ($\leq 3 \times 10^{-12}$), and the per-node gate reveals how the model allocates responsibility between the symmetric and interpretable pathways.

The exact equivariance guarantee. Unlike empirical equivariance (which is tested on specific rotations), the architectural guarantee holds for *all* rotations and *all* inputs simultaneously, by construction. This is possible because every scalar coefficient in the architecture is a function of rotation-invariant features only, and every vector is either a scalar $\times$ equivariant vector or an equivariant vector itself. The machine-precision verification (6×10^{-14} for gravity) confirms that the implementation matches the guarantee up to floating-point arithmetic.

Conservation as a regulariser. The physics-informed loss improves both conservation *and* MSE, suggesting that the conservation constraints constrain the optimisation landscape to physically plausible solutions. This is consistent with the broader finding that inductive biases improve sample efficiency [3], and suggests that conservation laws are particularly effective inductive biases for physical force-law learning.

Interpretable interaction graphs. The reasoning module’s edge weights provide an explicit, differentiable map from particle states to interaction strengths. The spring chain topology recall (0.615) and gravity strength correlation (0.283) demonstrate partial recovery; the full model distributes the force across both pathways, limiting individual interpretability. This is an honest result: the interaction graph is a harder target than acceleration prediction, and recovering it from end-to-end training without explicit supervision is a genuine open problem.

Limitations. (i) The experiments are at small scale (3–4 particles, 2D). Scaling to larger N requires replacing the fully-connected graph with a kNN or distance-cutoff graph, and GPU training for faster iteration. (ii) The interaction-graph recovery metrics are modest; explicit supervision on edge weights could improve them. (iii) The physics module assumes the potential function is known; extending to *discovering* the potential (as in Hamiltonian neural networks) is future work.

Reproducibility and broader impact

Reproducibility. All code, committed results, figures, and this paper’s source are in the repository (github.com/sehajr-singhs/physrnet). Every experiment is a self-contained script with fixed seeds. All experiments run on CPU in under three minutes per system. Installation: `pip install -r requirements.txt`.

Data availability. All datasets are generated synthetically within the codebase; no external data is required. The generation scripts are committed and deterministic (fixed seeds).

Code availability. The `physrnet` library is released under the MIT License. All modules, training loops, evaluation metrics, and figure-generation scripts are in the repository.

Broader impact. PhysRNet contributes to the growing body of work on physics-aware machine learning. The exact equivariance guarantee has potential applications in scientific computing, molecular dynamics, and robotics, where incorrect symmetry treatment can lead to unphysical predictions. The conservation-constrained training paradigm could improve long-horizon physical simulations. The interpretable interaction-graph module connects to the broader interpretability agenda in AI safety.

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